Lecture 16: Eigenspaces and Diagonalization

MATH203 Applied Linear Algebra

What Does P_i Project Onto?

Standing Assumption

Throughout this lecture:

$$\det(tI - A) = (t - \lambda_1)(t - \lambda_2)\cdots(t - \lambda_m), \qquad \lambda_i \text{ all distinct}$$

Same assumption as Lecture 15.

We have the spectral decomposition:

$$A = \lambda_1 P_1 + \lambda_2 P_2 + \cdots + \lambda_m P_m$$

where $P_1, \ldots, P_m$ are compatible projections that partition I.

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Today's question: What subspace does each P_i project **onto**?

A New Example

$$\underbrace{A = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix}}_{\text{our running example}}, \qquad \det(tI - A) = (t-1)(t-3)$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$.

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Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 3$.

Spectral projections (Lagrange formula from L15):

$$P_1 = \frac{A - 3I}{1-3} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}, \qquad P_3 = \frac{A - I}{3-1} = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix}$$

Value tables (from L15):

	$t = 1$	$t = 3$
A	1	3
P_1	**1**	0
P_3	0	**1**

$$A = \underbrace{1}_{\lambda_1} \cdot P_1 + \underbrace{3}_{\lambda_2} \cdot P_3$$

What Does P_i Do to a Vector?

Pick $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Apply each projection:

$$P_1\mathbf{v} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$P_3\mathbf{v} = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

What Does P_i Do to a Vector?

Pick $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$. Apply each projection:

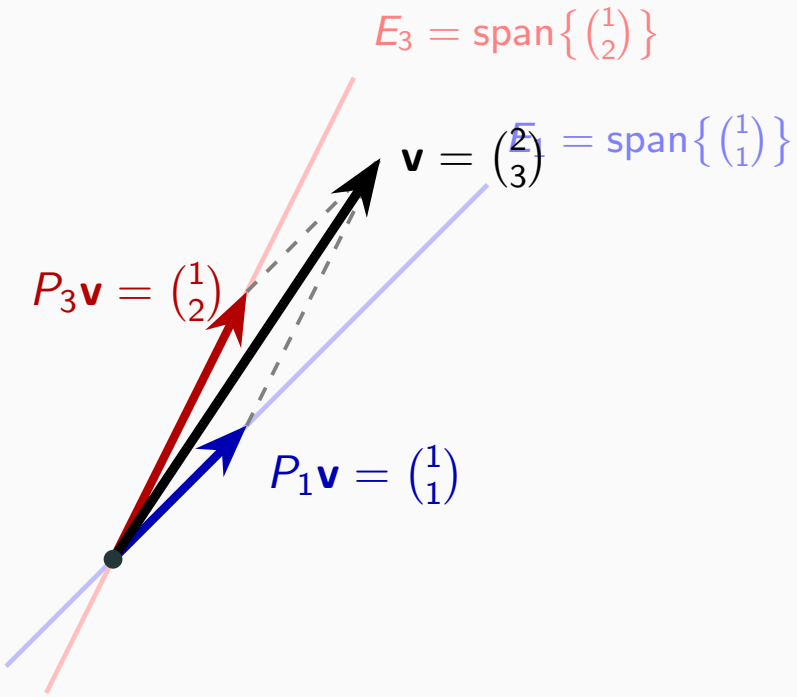
$$P_1\mathbf{v} = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$P_3\mathbf{v} = \begin{pmatrix} -1 & 1 \\ -2 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

Check: $\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \mathbf{v}$ ✓

Every Vector Splits Into Eigenspace Pieces

$$\mathbf{v} = P_1\mathbf{v} + P_3\mathbf{v} = \begin{pmatrix}1\\1\end{pmatrix} + \begin{pmatrix}1\\2\end{pmatrix} = \begin{pmatrix}2\\3\end{pmatrix} \checkmark$$



$P_1 + P_3 = I$: every vector decomposes into eigenspace pieces (parallelogram law).

What Does A Do to Each Piece?

Apply A to the first piece:

$$A \cdot P_1\mathbf{v} = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{1} \cdot P_1\mathbf{v}$$

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Apply A to the second piece:

$$A \cdot P_3\mathbf{v} = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = \mathbf{3} \cdot P_3\mathbf{v}$$

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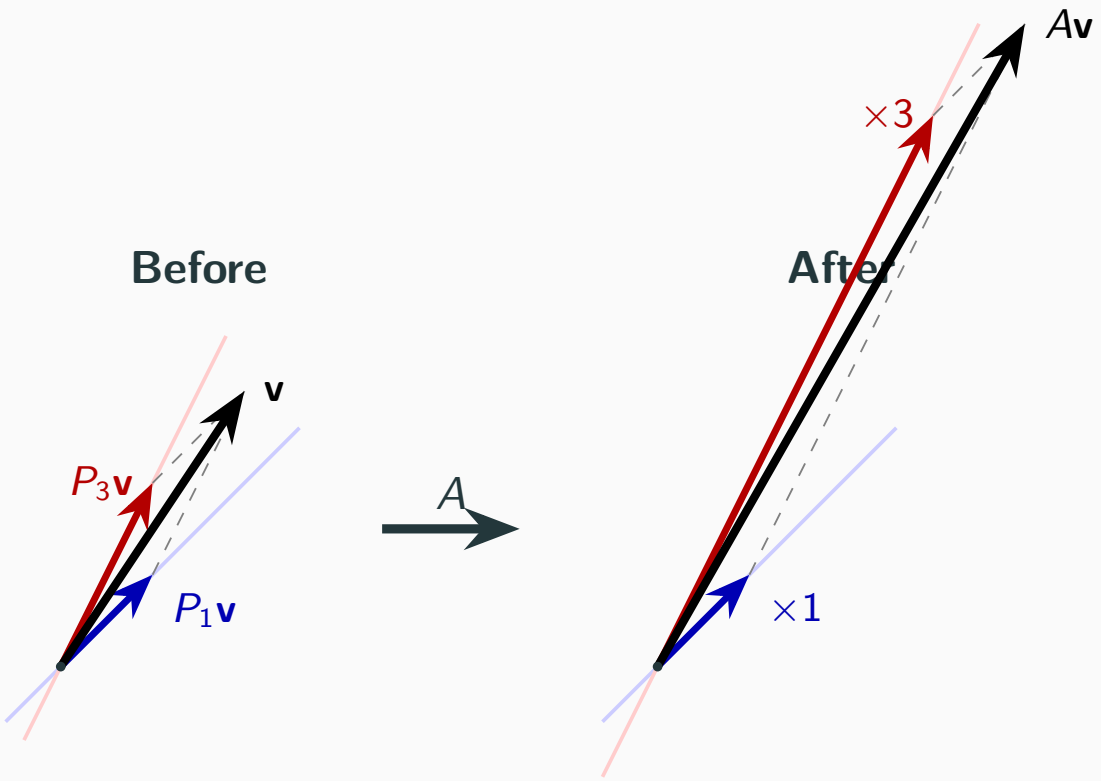
A **scales each piece by its eigenvalue**:
$\lambda_1 = 1$ and $\lambda_2 = 3$.

A Scales Each Piece by Its Eigenvalue

$$A\mathbf{v} = A\big(P_1\mathbf{v} + P_3\mathbf{v}\big) = \underbrace{1}_{\lambda_1}\cdot P_1\mathbf{v} + \underbrace{3}_{\lambda_2}\cdot P_3\mathbf{v}$$



A doesn't rotate or mix the pieces — it **only scales** each one along its eigenspace.

Why? — Value Table Proof

Why does A scale the P_1-piece by λ_1?

Multiply value tables entry-wise:

	$t = 1$	$t = 3$
A	1	3
$\times\ P_1$	1	0
AP_1	**1**	**0**

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Value table of $\lambda_1 P_1 = 1 \cdot P_1$:

	$t = 1$	$t = 3$
$1 \cdot P_1$	**1**	**0**

Same value table $\implies$ **same matrix**: $AP_1 = \lambda_1 P_1 = 1 \cdot P_1$.

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Similarly for P_3:

	$t=1$	$t=3$
A	1	3
$\times\ P_3$	0	1
AP_3	**0**	**3**

$=$ value table of $3P_3$ ✓

A Commutes With Each P_i

What about $P_1 A$? **Multiply** value tables in the other order:

	$t = 1$	$t = 3$
P_1	1	0
$\times\ A$	1	3
$P_1 A$	**1**	**0**

Same value table as AP_1! So $P_1 A = AP_1 = \lambda_1 P_1$.

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Value table multiplication is **entry-wise**: order doesn't matter.

$$AP_i = P_i A = \lambda_i P_i \qquad \text{for each } i$$

A commutes with every spectral projection.

(This fact will matter when we find **left eigenvectors** later.)

From $AP_i = \lambda_i P_i$ to Eigenvectors

Suppose $\mathbf{x} \in \mathrm{Col}(P_i)$.

Since P_i is a projection, it **fixes** its own column space: $P_i\mathbf{x} = \mathbf{x}$.

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Since P_i is a projection, it **fixes** its own column space: $P_i\mathbf{x} = \mathbf{x}$.

Then:

$$A\mathbf{x} = A\underbrace{P_i\mathbf{x}}_{=\mathbf{x}} = \underbrace{AP_i}_{=\lambda_i P_i}\mathbf{x} = \lambda_i \underbrace{P_i\mathbf{x}}_{=\mathbf{x}} = \lambda_i\mathbf{x}$$

Every vector in $\mathrm{Col}(P_i)$ is scaled by λ_i.

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Every vector in $\mathrm{Col}(P_i)$ is scaled by λ_i.

$\mathrm{Col}(P_i)$ = the subspace where A acts as

scalar multiplication by λ_i.

The Eigenspace

Definition (Eigenspace)

The **eigenspace** of A for eigenvalue λ_i is

$$E_{\lambda_i} \;=\; \mathrm{Col}(P_i) \;=\; \left\{ \mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \lambda_i \mathbf{x} \right\}$$

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Check on our example:

$$E_1 = \mathrm{Col}(P_1) = \mathrm{span}\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}. \quad A \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \mathbf{1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \checkmark$$

$$E_3 = \mathrm{Col}(P_3) = \mathrm{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}. \quad A \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} = \mathbf{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix} \checkmark$$

P_i projects onto E_{λ_i}:

the subspace where A acts as

scalar multiplication by λ_i.

Compatible Projections Revisited

Are P_1, P_3 Projections?

Check with value tables — **square each entry**:

	$t = 1$	$t = 3$
P_1	1	0
P_1^2	$1^2 = \mathbf{1}$	$0^2 = \mathbf{0}$

	$t = 1$	$t = 3$
P_3	0	1
P_3^2	$0^2 = \mathbf{0}$	$1^2 = \mathbf{1}$

Same value tables $\implies P_1^2 = P_1$ and $P_3^2 = P_3$. Both are **projections**. ✓

Compatible? Partition of /?

Compatible? Multiply value tables entry-wise:

	$t = 1$	$t = 3$
P_1	1	0
$\times\ P_3$	0	1
$P_1 P_3$	**0**	**0**

Value table all zeros $\implies P_1 P_3 = 0$. **Compatible!** ✓

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Partition of I? Add value tables:

	$t = 1$	$t = 3$
P_1	1	0
$+\ P_3$	0	1
$P_1 + P_3$	**1**	**1**

Value table = value table of I. $P_1 + P_3 = I$. ✓

Connection to Lecture 8

From Lecture 8: compatible projections that partition I decompose every vector **uniquely**:

$$\mathbf{v} = P_1\mathbf{v} + P_3\mathbf{v}, \qquad P_1\mathbf{v} \in E_1, \quad P_3\mathbf{v} \in E_3$$

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The Lagrange formula from L15 **automatically**

produces compatible projections.

No extra verification needed — it's built into the construction.

Eigenvectors With Different Eigenvalues Are Independent

Theorem

Let $\mathbf{v}_1, \ldots, \mathbf{v}_m$ be eigenvectors of A with **distinct** eigenvalues $\lambda_1, \ldots, \lambda_m$. Then $\mathbf{v}_1, \ldots, \mathbf{v}_m$ are **linearly independent**.

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Proof: Suppose $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$.

Apply P_i to both sides. Since $\mathbf{v}_j \in E_{\lambda_j}$ and $P_iP_j = 0$ for $i \neq j$:

$$a_1 \underbrace{P_i\mathbf{v}_1}_{=0} + \cdots + a_i \underbrace{P_i\mathbf{v}_i}_{=\mathbf{v}_i} + \cdots + a_m \underbrace{P_i\mathbf{v}_m}_{=0} = \mathbf{0}$$

So $a_i\mathbf{v}_i = \mathbf{0}$. Since $\mathbf{v}_i \neq \mathbf{0}$ (eigenvector), we get $a_i = 0$.

This holds for **every** i. Therefore $a_1 = a_2 = \cdots = a_m = 0$. $\square$

Unique Decomposition (Lecture 9 Callback)

From Lecture 9: compatible projections that partition I give a **unique decomposition** of every vector.

Since $P_1 + P_3 = I$, every vector in $\mathbb{R}^n$ decomposes as:

$$\mathbf{v} = \underbrace{P_1\mathbf{v}}_{\in E_1} + \underbrace{P_3\mathbf{v}}_{\in E_3}$$

By the Proposition (linear independence), this decomposition is **unique**: there is only one way to write $\mathbf{v}$ as a sum of eigenvectors from each eigenspace.

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By the Proposition (linear independence), this decomposition is **unique**: there is only one way to write $\mathbf{v}$ as a sum of eigenvectors from each eigenspace.

The eigenspaces $E_{\lambda_1}, \ldots, E_{\lambda_m}$ together **span** all of $\mathbb{R}^n$.

Every vector decomposes **uniquely** into eigenvector pieces.

(This is the geometric meaning of $P_1 + P_2 + \cdots + P_m = I$.)

Cross-Filling the Spectral Projections

A New Example — Repeated Eigenvalue

$$\underbrace{B = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix}}_{\text{eigenvalues } 2,2,5}, \qquad \det(tI - B) = (t-2)^2(t-5)$$

Two **distinct** eigenvalues: $\lambda_1 = 2$, $\lambda_2 = 5$.

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Spectral projections (Lagrange formula):

$$P_5 = \frac{B - 2I}{5-2} = \frac{1}{3}\begin{pmatrix} 0 & 0 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$P_2 = I - P_5 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

	$t = 2$	$t = 5$
B	2	5
P_2	**1**	0
P_5	0	**1**

P_2 Has Rank 2 — A Plane

$$P_2 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \qquad \text{rank} = 2$$

$E_2 = \text{Col}(P_2)$ is a **plane** in $\mathbb{R}^3$, not just a line.

$$P_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \qquad \text{rank} = 1$$

$E_5 = \text{Col}(P_5)$ is a **line** in $\mathbb{R}^3$.

P_2 Has Rank 2 — A Plane

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$$P_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \qquad \text{rank} = 1$$

$E_5 = \text{Col}(P_5)$ is a **line** in $\mathbb{R}^3$.

To find eigenvectors, we **cross-fill** each projection (Lecture 8).

Cross-Fill P_2 — Step 1

$$P_2 = \begin{pmatrix} \boxed{1} & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Pivot at position $(1,1)$, value $= 1$.

Column $\div$ pivot: $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. Row: $\begin{pmatrix} 1 & 0 & -1 \end{pmatrix}$.

Rank-1 piece:

$$R_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Remainder: $P_2 - R_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

Cross-Fill P_2 — Step 2 and Assembly

Remainder: $\begin{pmatrix} 0 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}$. Pivot at $(2,2)$, value $= 1$.

Column $\div$ pivot: $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. Row: $\begin{pmatrix} 0 & 1 & -1 \end{pmatrix}$.

Cross-Fill P_2 — Step 2 and Assembly

Remainder: $\begin{pmatrix} 0 & 0 & 0 \\ 0 & \boxed{1} & -1 \\ 0 & 0 & 0 \end{pmatrix}$. Pivot at $(2,2)$, value $= 1$.

Column $\div$ pivot: $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$. Row: $\begin{pmatrix} 0 & 1 & -1 \end{pmatrix}$.

Two-factor form:

$$P_2 = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}}_{U_2\ (3\times 2)} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix}}_{V_2\ (2\times 3)}$$

Check: $V_2 U_2 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$ ✓

Cross-Fill P_5 — One Step

$$P_5 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & \boxed{1} \end{pmatrix}$$

Pivot at $(3,3)$, value $= 1$.

Column $\div$ pivot: $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Row: $\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$.

$$P_5 = \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{U_5} \underbrace{\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}}_{V_5}$$

Check: $V_5 U_5 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1$ ✓

Columns = Right Eigenvectors

Columns of U_2: $\begin{pmatrix}1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}$. Column of U_5: $\begin{pmatrix}1\\1\\1\end{pmatrix}$.

Verify $B\mathbf{u} = \lambda\mathbf{u}$:

$$B\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}2\\0\\0\end{pmatrix} = \mathbf{2}\begin{pmatrix}1\\0\\0\end{pmatrix} \checkmark \qquad B\begin{pmatrix}0\\1\\0\end{pmatrix} = \begin{pmatrix}0\\2\\0\end{pmatrix} = \mathbf{2}\begin{pmatrix}0\\1\\0\end{pmatrix} \checkmark$$

$$B\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}5\\5\\5\end{pmatrix} = \mathbf{5}\begin{pmatrix}1\\1\\1\end{pmatrix} \checkmark$$

The columns of U_i live in $\mathrm{Col}(P_i) = E_{\lambda_i}$.

Rows = Left Eigenvectors

Rows of V_2: $\begin{pmatrix} 1 & 0 & -1 \end{pmatrix}$, $\begin{pmatrix} 0 & 1 & -1 \end{pmatrix}$. Row of V_5: $\begin{pmatrix} 0 & 0 & 1 \end{pmatrix}$.

Verify $\mathbf{v}^T B = \lambda \mathbf{v}^T$:

$$\begin{pmatrix} 1 & 0 & -1 \end{pmatrix} B = \begin{pmatrix} 2 & 0 & -2 \end{pmatrix} = \mathbf{2} \begin{pmatrix} 1 & 0 & -1 \end{pmatrix} \checkmark$$

$$\begin{pmatrix} 0 & 1 & -1 \end{pmatrix} B = \begin{pmatrix} 0 & 2 & -2 \end{pmatrix} = \mathbf{2} \begin{pmatrix} 0 & 1 & -1 \end{pmatrix} \checkmark$$

$$\begin{pmatrix} 0 & 0 & 1 \end{pmatrix} B = \begin{pmatrix} 0 & 0 & 5 \end{pmatrix} = \mathbf{5} \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \checkmark$$

Left eigenvectors "detect" from the left: $\mathbf{v}^T B = \lambda \mathbf{v}^T$.

Cross-Eigenspace Orthogonality

Different eigenspaces ignore each other:

$$V_2 U_5 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$V_5 U_2 = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \end{pmatrix}$$

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Why? From $P_2 P_5 = 0$:

$$\underbrace{U_2 V_2}_{P_2} \cdot \underbrace{U_5 V_5}_{P_5} = U_2 \, (\underbrace{V_2 U_5}_{=0}) \, V_5 = 0$$

Left eigenvectors "recognize" their own right eigenvectors ($V_i U_i = I$) and "ignore" the others ($V_i U_j = 0$).

Cross-filling $P_i = U_i V_i$ gives:

Columns of U_i = **right eigenvectors** ($B\mathbf{u} = \lambda_i \mathbf{u}$)

Rows of V_i = **left eigenvectors** ($\mathbf{v}^T B = \lambda_i \mathbf{v}^T$)

Within: $V_i U_i = I$ Across: $V_i U_j = 0$

Diagonalization: Stacking the Cross-Fillings

Stack Everything Together

Right eigenvector matrix — stack columns:

$$\underbrace{P}_{3\times 3} = \begin{pmatrix} U_2 & U_5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

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Left eigenvector matrix — stack rows:

$$\underbrace{Q}_{3\times 3} = \begin{pmatrix} V_2 \\ V_5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

$Q = P^{-1}$ — For Free!

Multiply QP block by block:

$$QP = \begin{pmatrix} V_2 \\ V_5 \end{pmatrix} \begin{pmatrix} U_2 & U_5 \end{pmatrix} = \begin{pmatrix} V_2 U_2 & V_2 U_5 \\ V_5 U_2 & V_5 U_5 \end{pmatrix} = \begin{pmatrix} I_2 & 0 \\ 0 & 1 \end{pmatrix} = I_3$$

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$$QP = \begin{pmatrix} V_2 \\ V_5 \end{pmatrix} \begin{pmatrix} U_2 & U_5 \end{pmatrix} = \begin{pmatrix} V_2 U_2 & V_2 U_5 \\ V_5 U_2 & V_5 U_5 \end{pmatrix} = \begin{pmatrix} I_2 & 0 \\ 0 & 1 \end{pmatrix} = I_3$$

So $Q = P^{-1}$. We get P^{-1} **without computing any inverse**!

$$\underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}_{P^{-1}} \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{P} = I_3 \checkmark$$

The Diagonalization Formula

$$B = 2\underbrace{U_2 V_2}_{P_2} + 5\underbrace{U_5 V_5}_{P_5}$$

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Factor:

$$B = \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{P} \underbrace{\begin{pmatrix} 2 & & \\ & 2 & \\ & & 5 \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}}_{P^{-1}}$$

Verify:

$$PD = \begin{pmatrix} 2 & 0 & 5 \\ 0 & 2 & 5 \\ 0 & 0 & 5 \end{pmatrix}, \quad PDP^{-1} = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix} = B \ \checkmark$$

The Cancellation Trick

What is B^2?

$$B^2 = (\overbrace{PDP^{-1}}^{B})(\overbrace{PDP^{-1}}^{B}) = PD\underbrace{\cancel{P^{-1}P}}_{=I}DP^{-1} = PD^2P^{-1}$$

The $P^{-1}P$ in the middle **cancels**!

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Powers:

$$B^n = PD^nP^{-1}, \qquad D^n = \begin{pmatrix} 2^n & & \\ & 2^n & \\ & & 5^n \end{pmatrix}$$

Diagonal matrices are trivial to exponentiate.

Any Polynomial — Same Trick

For **any** polynomial g:

$$g(B) = P\, g(D)\, P^{-1}, \qquad g(D) = \begin{pmatrix} g(2) & & \\ & g(2) & \\ & & g(5) \end{pmatrix}$$

All the middle terms cancel. Computing g on a diagonal matrix = just plug in eigenvalues.

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Connection to value tables:

	$t = 2$	$t = 5$
$g(B)$	$g(2)$	$g(5)$

Spectral formula: $g(B) = g(2)\,P_2 + g(5)\,P_5$.

Diagonalization: $g(B) = P\,g(D)\,P^{-1}$.

Same machine, two notations. The cancellation algebra is **why** value tables work.

Eigenspace Geometry: The Dominant Eigenvalue

A^n via Value Tables

Back to our 2×2 example: $A = \begin{pmatrix} -1 & 2 \\ -4 & 5 \end{pmatrix}$, eigenvalues $1, 3$.

	$t = 1$	$t = 3$
A	1	3
A^2	1	9
A^3	1	27
A^n	1	3^n

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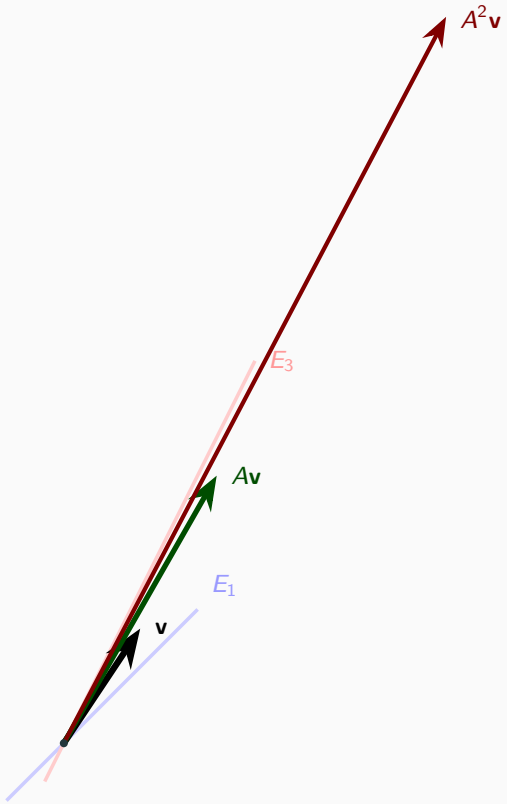
For $\mathbf{v} = \binom{2}{3}$: $\quad A^n\mathbf{v} = \underbrace{1^n}_{=1} \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{P_1\mathbf{v}} + \underbrace{3^n}_{\to\infty} \underbrace{\begin{pmatrix} 1 \\ 2 \end{pmatrix}}_{P_3\mathbf{v}}$

As $n \to \infty$: the E_3 piece grows like 3^n, the E_1 piece stays $= 1$.

The dominant eigenvalue wins: $A^n\mathbf{v}$ aligns with $E_3 = \operatorname{span}\left\{ \binom{1}{2} \right\}$.

Visualizing A^n



Each step: the E_3 piece stretches $\times 3$, the E_1 piece is unchanged.

Eigenvalues tell you **how much** each direction stretches.

Eigenspaces tell you **which** directions.

Together, they completely describe what A does.

Our Restriction: The Minimal Polynomial

A Repeated Root That Worked

B from §3 had $\det(tI - B) = (t-2)^2(t-5)$ — a **repeated** root!

Yet the spectral machinery worked perfectly:

- Two distinct projections P_2, P_5 ✓
- Compatible, partition of I ✓
- Cross-filling, eigenvectors ✓
- $B = PDP^{-1}$ ✓

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- $B = PDP^{-1}$ ✓

Why? The characteristic polynomial can repeat. What matters is a different polynomial.

The Minimal Polynomial

Definition (Minimal Polynomial)

The **minimal polynomial** $m(t)$ is the lowest-degree monic polynomial such that $m(A) = 0$.

It divides every annihilating polynomial (including $\det(tI - A)$).

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Check on B:

$$(B - 2I)(B - 5I) = \begin{pmatrix} 0 & 0 & 3 \\ 0 & 0 & 3 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} -3 & 0 & 3 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

So $m(t) = (t - 2)(t - 5)$ — **distinct** linear factors. ✓

Another Fine Example: $A = 3I$

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}, \qquad \det(tI - A) = (t-3)^2 \quad \text{(repeated!)}$$

But $A - 3I = 0$ already, so $m(t) = t - 3$ (degree 1, one distinct factor).

Spectral decomposition: $A = 3 \cdot I$. Just one projection $P_1 = I$.

The whole space $\mathbb{R}^2$ is one eigenspace. Fine!

Beyond Our Assumption: The Shear

$$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, \qquad \det(tI - J) = (t-2)^2$$

$$J - 2I = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0. \qquad (J-2I)^2 = 0.$$

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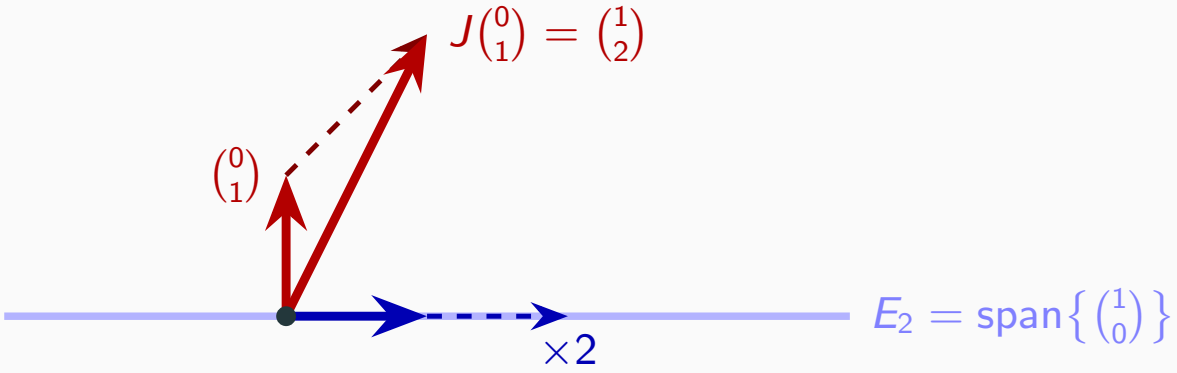

Vectors on E_2 scale by 2. Vectors off E_2 get **sheared** — not just scaled.

This situation needs **new tools** — a topic for a future lecture.

The Real Restriction

The characteristic polynomial CAN have repeated roots.

The **minimal polynomial** is what matters:

Distinct linear factors in $m(t) \implies$ **diagonalizable**.

Repeated factor in $m(t) \implies$ needs new tools (future lecture).

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	Char. poly	Min. poly	
$B = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix}$	$(t-2)^2(t-5)$	$(t-2)(t-5)$	✓
$3I_2$	$(t-3)^2$	$(t-3)$	✓
$J = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$	$(t-2)^2$	$(t-2)^2$	×

Summary

1. $E_{\lambda_i} = \mathrm{Col}(P_i)$: eigenspace = column space of spectral projection.
2. Compatible projections (value tables verify instantly): eigenvectors with different eigenvalues are independent; every vector decomposes uniquely.
3. Cross-fill $P_i = U_i V_i$: columns of U_i = right eigenvectors, rows of V_i = left eigenvectors. $V_i U_j = \delta_{ij} I$.
4. Stack: $A = PDP^{-1}, \quad g(A) = Pg(D)P^{-1}$ reduces every computation to plugging eigenvalues into g.
5. The **minimal polynomial** is the real restriction: distinct linear factors $\implies$ diagonalizable.

Next Lecture

We can compute A^n via eigenvalues. But what about e^A?

$$e^A = \lim_{n\to\infty} \left(I + \frac{A}{n} \right)^n$$

Next: The exponential function, matrix exponential, and differential equations.